Operations on sets

1 Explain the meaning of the following statements:

a Complement of set A.

b Union of set A and set B.

c Intersection of set A and set B.

d Difference of two sets A and B.

2 Suppose set $A = \{3, 4, 5, 6, 7\}$ and set $B = \{3, 7, 8, 9\}$. Find $A \cap B$.

3 List the elements of $A \cap B$ given the following sets:

a $A = \{\text{even numbers}\}$ and $B = \{\text{square numbers less than } 100\}$

b $A = \{\text{odd numbers}\}$ and $B = \{\text{factors of } 105\}$

c $A = \{\text{multiples of } 5\}$ and $B = \{\text{positive numbers less than } 50\}$

4 Set A is the set of possible outcomes from rolling a standard die, and set B is the set of possible outcomes from rolling an eight-sided die. List the elements of the following sets:

a A

b B

c $A \cap B$

d $A \cup B$

5 For the following sets, list the elements of:

i $B \cup A$

ii $A \cap B$

a A is the set of factors of 24, and B is the set of factors of 36.

b A is the set of prime numbers less than 40 and B is the set of positive integers less than 12.

6 For the following sets, describe the following:

i $A \cup A'$

ii $A' \cap B'$

a Let A be the set factors of 24, and B be the set factors of 36.

b Let A be the set multiples of 4 less than 70, and B be the set multiples of 6 less than 70.

7 Let A be the set of multiples of 4 less than 70, and B be the set of multiples of 6 less than 70.

a Describe $(A \cap B)'$.

b Describe $A' \cap B'$.

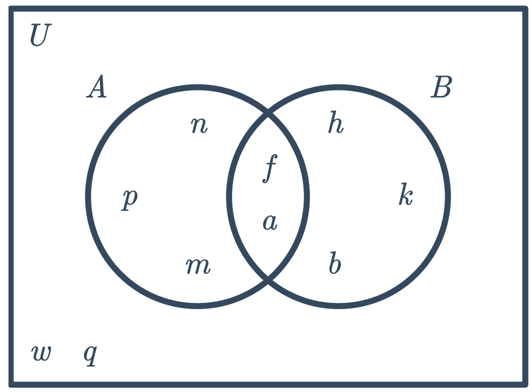
c Is $A' \cap B'$ the same as $(A \cap B)'$?

8 Let A be the set of multiples of 4 less than 70, and B be the set of multiples of 6 less than 70.

- **a** Describe $(A \cup B)'$.
- **b** Describe $A' \cup B'$.
- **c** Is $(A \cup B)'$ the same as $A' \cup B'$?

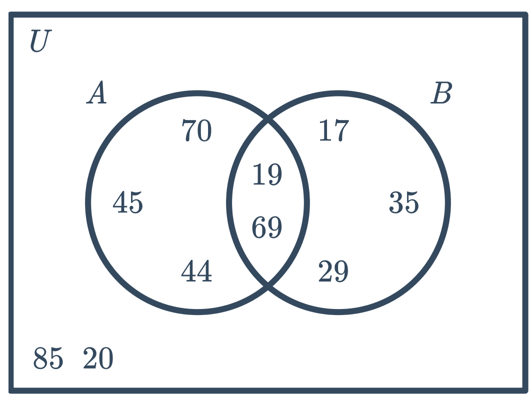
9 Consider the following Venn diagram:

- **a** Find the set $A - B$.
- **b** Find the set $A - B'$.



10 Consider the following Venn diagram:

- **a** Find the set $A' - B$.
- **b** Find the set $(A - B)'$.



Complementary events

11 Determine whether the following pairs of events are complementary or not:

a Event A: Selecting a positive number.
Event B: Selecting a negative number.

b Event A: Drawing a red card from a standard deck of cards (no jokers).
Event B: Drawing a black card from a standard deck of cards (no jokers).

c Event A: Drawing a club from a standard deck of cards (no jokers).
Event B: Drawing a spade from a standard deck of cards (no jokers).

d Event A: Rolling a number greater than 3 on a die.
Event B: Rolling a number less than 3 on a die.

12 Find each of the following probabilities, find the probablity that the complementary event will occur:

a $\dfrac{4}{5}$

b 0.64

13 A biased coin is flipped, with heads and tails as possible outcomes. Calculate $P\left(\text{heads}\right)$ if $P\left(\text{tails}\right) = 0.56$.

14 A bag contains 34 red marbles and 35 blue marbles. If picking a marble at random, find:

a $P\left(\text{red}\right)$

b $P\left(\text{not red}\right)$

15 A bag contains 50 black marbles, 37 orange marbles, 29 green marbles and 23 pink marbles. If a marble is selected at random, find the following:

a $P\left(\text{orange}\right)$

b $P\left(\text{orange or pink}\right)$

c $P\left(\text{not orange}\right)$

d $P\left(\text{neither orange nor pink}\right)$

16 A number between 1 and 100 inclusive is randomly picked.

a State the complement of picking a number greater than 61.

b Find the probability that the number picked is greater than 61.

17 A regular die is rolled. Find the probability of:

a Not rolling a 4.

b Not rolling a 1 or 5.

c Not rolling an even number.

d Not rolling an 8.

e Not rolling a 1, 2, 3, 4, 5, or 6.

18 The 26 letters of the alphabet are written on pieces of paper and placed in a bag. If one letter is picked out of the bag at random, find the probability of:

a Not selecting a B.

b Not selecting a K, R or T.

c Selecting a letter that is not in the word PROBABILITY.

d Not selecting a T, L, Q, A, K or Z.

e Selecting a letter that is not in the word WORKBOOK.

19 A card is drawn at random from a standard deck. Find the probability that the card is:

a A diamond.

b A spade.

c Not a heart.

20 From a normal deck of cards, find the probability of:

a Selecting a five.

b Selecting a nine.

c Not selecting a two.

d Selecting a black card.

e Not selecting a black card.

21 A bag contains 86 marbles, some of them are black and some are white. If the probability of selecting a black marble is $\frac{33}{43}$, find:

a The number of black marbles.

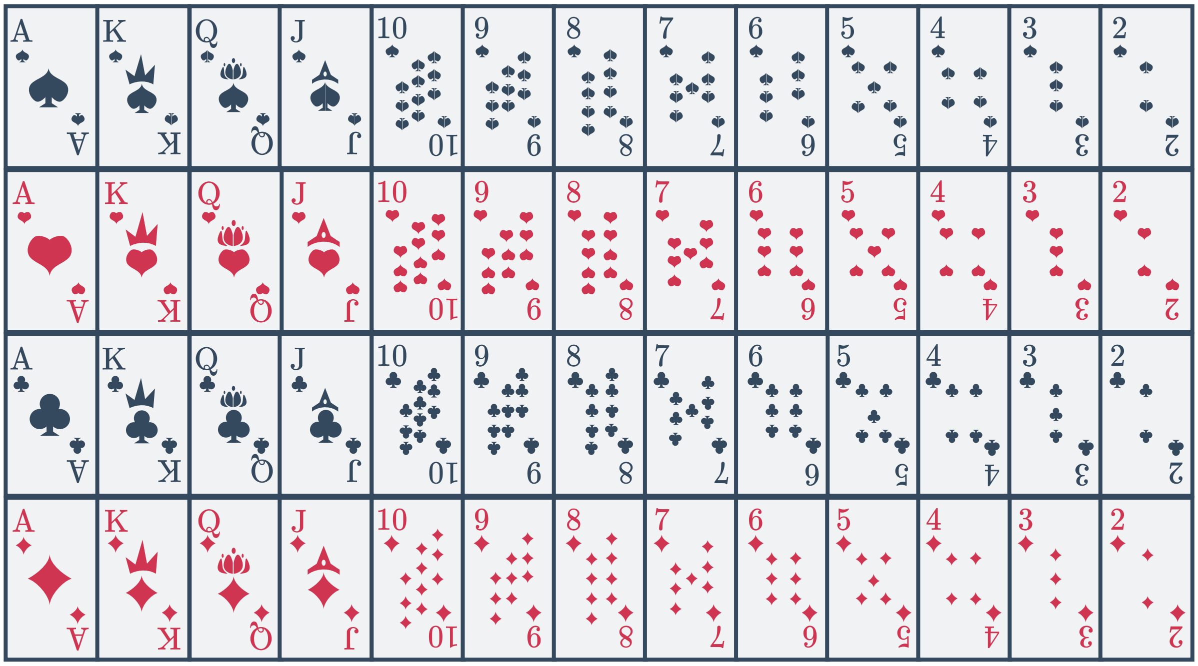
b The number of white marbles.

22 The number of movie, concert and music tickets on offer as a prize are in the ratio $8 : 19 : 3$.

a Find the probability that the winner will be given a concert ticket.

b The winner doesn't want to see a musical. Find the probability that they get a ticket they do want.

23 A card is selected from a standard deck of c[redacted]



Find the probability of:

a Selecting a face card.

b Selecting a black nine.

c Selecting an odd-numbered black card, not counting ace as a numbered card.

d Selecting a red nine.

e Not selecting a red three.

f Not selecting a queen of clubs.

g Selecting a ten, jack, queen, king, or ace.

h Selecting the king of diamonds.

i Not selecting a red ten or black jack.

24 A circular spinner is divided into unequal parts. The green sector takes up an angle of 250° at the center. The red sector takes up an angle of 60° at the center and the blue sector takes up the remainder of the spinner.
Find the probability that the spinner will land on blue.

Probabilities from Venn diagrams

25 We are interested in the color of a card randomly drawn from a standard deck. Draw a Venn

26 Consider the Venn diagram:
Find:

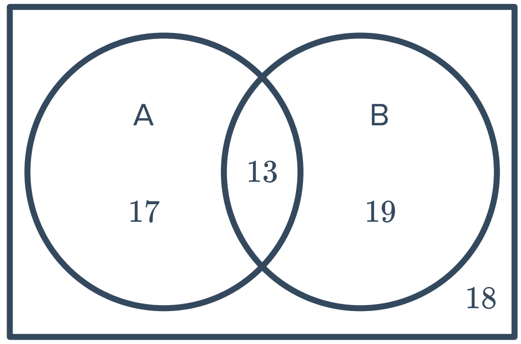
a $P(A)$

b $P(B)$

c $P(\text{not } A)$

d $P(\text{not } B)$

e $P(B \text{ only})$



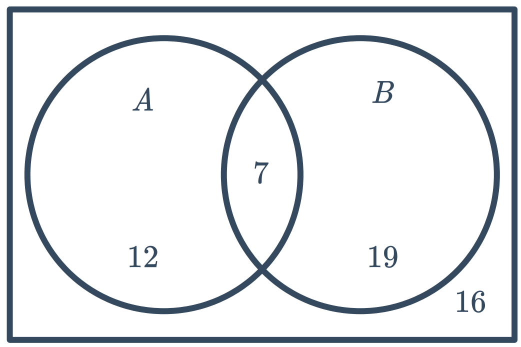
27 Consider the given Venn diagram:
Find:

a $P(A \text{ but not } B)$

b $P(A \text{ and } B)$

c $P(A \text{ or } B)$

d $P(\text{neither } A \text{ nor } B)$



28 Some people were asked what form of exercise they do. The results are displayed in the Venn diagram below:

a State whether the following are correct:

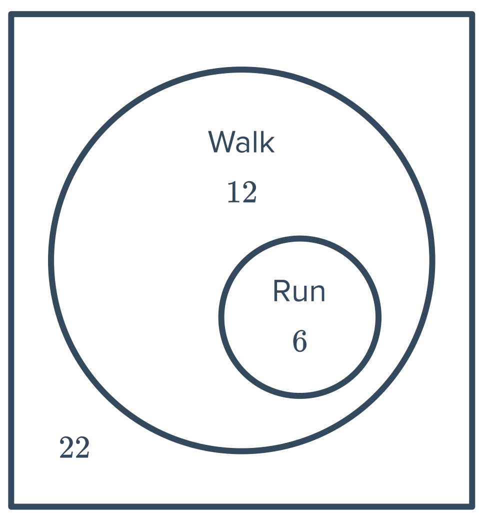
i 12 people walk only.

ii 22 people run only.

iii 6 people run only.

iv 6 people walk only.

b If one person is chosen at random, find the probability that they walk for exercise.



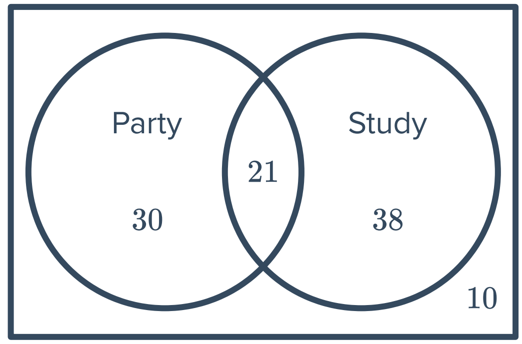
29 The Venn diagram shows the number of students choosing to study on the night before an exam, and the number of students choosing to party:
Find the probability that a student selected at random chose:

a Not to party.

b To study and party.

c To study only.

d To study or party.



30 The Venn diagram shows the decisions of 448 workers to either work or strike on a particular day of industrial action.
Find the probability that a worker selected randomly chose to:

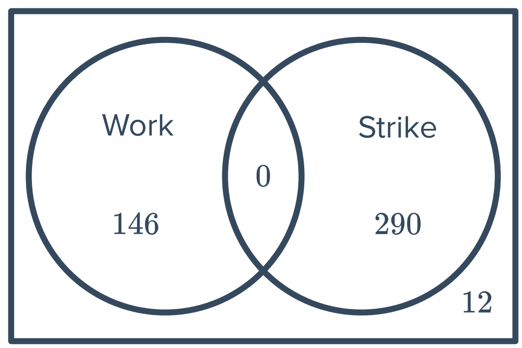
a Strike.

b Strike and work.

c Work and not strike.

d Work or strike.

e Neither work nor strike.



31 The Venn diagram shows the decisions of 535 consumers choosing to buy an iPhone and consumers choosing to buy a Blackberry:
If a consumer is selected at random, find the probability that he chose to buy:

a A Blackberry.

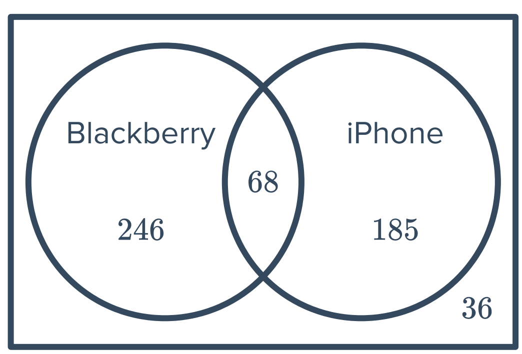
b A Blackberry only.

c Both phones.

d An iPhone or a Blackberry.

e Neither phones.

f An iPhone but not a Blackberry.



32 In a music school of 129 students, 83 studen[redacted]y the piano, 80 students play the guitar and 14 students play neither.

a Construct a Venn diagram for this situation.

b Find the probability that a student chosen at random plays:

i Both the piano and the guitar.

ii The piano or the guitar.

iii Neither the piano nor the guitar.

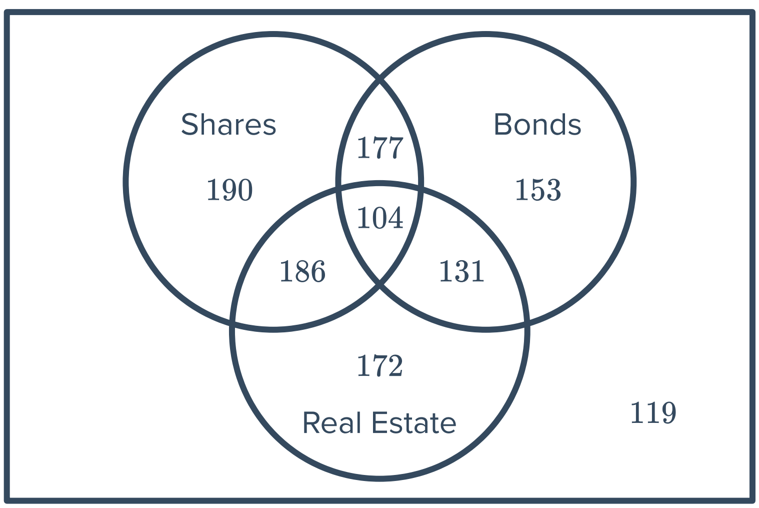
33 A grade of 172 students are to choose to study either Mandarin or Spanish (or both). 79 students choose Mandarin and 111 students choose Spanish.

a How many students have chosen both languages?

b If a student is picked at random, find the probability that the student has chosen Spanish only.

c If a student is picked at random, find the probability that the student has not chosen Mandarin.

34 The Venn diagram depicts the investment choices of 1232 investors:



Find the probability that an investor randomly selected has investments in:

a Bonds.

b Bonds and real estate.

c Bonds or real estate.

d Bonds and real estate but not shares.

e Shares, bonds and real estate.

f Real estate and shares but not bonds.

g Shares, bonds or real estate.

35 In a survey of 31 students, it was found that:

- 16 students play tennis.
- 14 students play hockey.
- 2 students play none of these sports.
- 8 play both tennis and cricket.
- 7 play both cricket and hockey.
- 6 play both tennis and hockey.
- 3 play all three.

a Construct a Venn diagram for this situation.

b Find the probability that a randomly selected student plays all three sports.